

The U.S. and China: Going Head-to-Head on AI in 2025 - SM

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Executive Summary

The close of 2024 and start of 2025 witnessed four remarkable AI developments that caught the attention of industry analysts, policymakers, and economists alike—several of which would have been startling in any other era or time of year.

The American AI industry was on the receiving end of the commercial equivalent of a “Sputnik moment”. China’s DeepSeek released an open-source LLM so effective and efficient that it ranked near the top of the leaderboards in performance—but at a cost of mere pennies on the dollar relative to OpenAI’s offerings.¹

President Donald Trump signed the nation’s first-ever declaration of a nationwide energy emergency, in part to answer the pleas of AI industry executives for more energy to support their compute needs.²

Outgoing President Joe Biden signed an all-but-ignored executive order directing two of the largest government agencies in the world, the U.S. Department of Defense and U.S. Department of Energy, to make their respective real estate holdings available for leases in order to expedite the construction or expansion of large-scale private sector data centers and their accompanying transmission lines.³ DOE has authority over 2.8 million acres of land holdings, the vast majority of which is owned directly.⁴ DOD has oversees and manages 30 million acres.⁵ If counted as a U.S. state, DOD and DOE acreage combined would measure about 51,000 square miles and rank larger than 23 states.⁶

President Trump and several industry leaders announced Stargate, a \$500 billion public-private collaboration—including, at the outset, OpenAI, Softbank, and Oracle—for investment in AI infrastructure. While \$500 billion is a headline grabbing figure, it is \$500 billion over 4 years, which is not large relative to the needs of the industry. Microsoft alone, for example, is investing \$80 billion in 2025. AWS, Microsoft, Alphabet and Meta spent \$113 billion in capex in 2024, and Oracle, spent \$7 billion.

Goldman Sachs forecasts that capex by the big four in 2025 and 2026 alone will be \$560 billion, with \$279 billion of that in 2025. Another \$720 billion of grid capex (transmission and distribution) will need to be invested to support the pipeline of data center projects that are currently in flight.⁷

Rarely in the history of American commerce has an industry vaulted in the span of a few months from a largely pioneering phase past the go-to-market phase and onward to potentially hitting the wall with respect to basic factors of production, technological capabilities, and the ability of mature domestic infrastructure systems to respond.

Yet, in the span of roughly one year following the release of widely OpenAI's generative intelligence model, the artificial intelligence industry began encountering occasional brakes and caution-ahead signs with respect to core technology (e.g., chips⁸), factors of production (electricity to data centers) and systems-level infrastructure (e.g., transformers in electrical utilities⁹), with red flashing lights on the horizon.

This rapid evolution has had four consequences:

- **Wide Variance of Forecasts.** Industry analysts assessing both the artificial intelligence sector and the energy sector have had difficulty converging on supply and demand forecasts.

America's AI and energy sectors have melded together, and such interdependence will be largely irreversible.

- **Melding.** In the United States, the fate of artificial intelligence and the energy sector can no longer be unbundled. One cannot speak authoritatively about one without incorporating the interests and issues of the other: interdependence is here to stay. Like it or not, America now has an AI-energy complex.
- **China Ascendant.** While the U.S. holds a commanding lead in areas such as research, frontier models, and private sector financing, China has been able to close the technological gap on nearly all other fronts—additional energy generation, in particular, is a marginal worry for China—and it is highly likely that, as of 2025, China will match or surpass the United States in its degree of control over its AI destiny writ large.
- **Policy Priorities.** The interdependence of the U.S. artificial intelligence and energy sectors, fluid levels of potential subsidies and actual disbursements,

concerns over the degree of government intervention and procedural fairness, and fitful progress in permitting and location of energy generation and transmission additions will make it increasingly difficult for policymakers at all levels to determine which policy interventions deserve the highest priority.

This paper has several aims: (1) put the recent AI industry developments in context; (2) clarify to some extent where U.S. industry is, and is not, hitting the wall; (3) highlight the key determinants of the energy-AI interdependence; (4) bring the degree of China's ascent into sharp relief; and (5) map policy reforms that legislators and regulators may want to consider. Investment opportunities will be covered in a follow up paper.

This analysis finds that energy interdependence is not the sole rate-limiting factor facing U.S. industry.

However, it is by far, the near, mid- and long-term constraint that will have the most far-reaching consequences. For that reason, China is highly likely to match or surpass the United States' degree of autonomy in artificial intelligence this year or soon after, as measured by practicality, affordability, distribution, and reach into global markets

In AI, China is soon highly likely to have more control over its destiny than the U.S. has over its own.

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Overview

Rarely in the history of global industry and commerce has an industry vaulted in the span of months from a predominantly pioneering phase past the go-to-market phase and onward to potentially hitting the wall with respect to basic factors of production, core technological components, and the ability of mature infrastructure systems to respond.

Even in the heyday of the early booming automobile industry at the start of the 20th century, when thousands of small manufacturers flooded the marketplace, a decade or so passed before rubber supply became a worry and steel was not a concern until several years after that. There is little evidence that a reliable electricity supply has ever been a paramount constraint for American automobile manufacturers.

Yet this rapid acceleration into growth-rate limits is what we have just witnessed since the advent of generative artificial intelligence, and the realization of constraints came in roughly the span of one year.

AI, prior to the release of ChatGPT in late 2022, was regarded as either applicable to narrow, practical use cases (e.g., credit scoring, social media algorithms, cruise control, etc.) or highly speculative in nature (artificial general intelligence, artificial superintelligence).

Now generative and predictive AI are seemingly everywhere, in use or being considered for ever more use cases in nearly every industry, and practical boundaries to scaling have come into view: the so-called data wall, the availability of basic factors of production such as rare earth minerals, the availability of semiconductors, the capacity of data centers, and—last and increasingly worrisome for America—the ability of energy generation facilities and grids to respond to rapidly growing demand in the timeframe expected by those leading the way in AI.

The consensus among industry and government analysts to date has been that, writ large, the United States would sustain its broad, commanding lead over its nearest competitor, China, in AI research, development, infrastructure, and end-user applications through the end of this decade, if not longer.

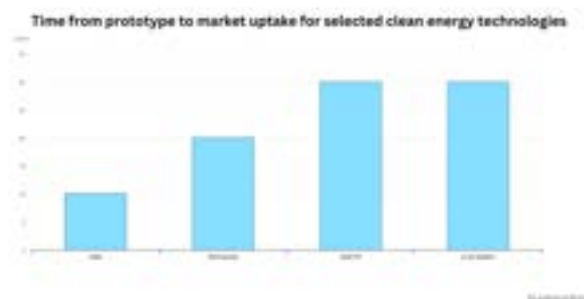
This assertion is no longer supported by a rigorous analysis of open-source information and data, a reevaluation of key metrics, and reasonable inferences regarding compute requirements for the next generation of LLMs. As one industry observer noted wryly at the close of last year: 2024 was the year when the U.S. AI industry had a rude awakening to the reality of America's energy sector, and the U.S. energy sector had a rude awakening to the reality of AI.¹⁰

Looking ahead, 2025 will be the year that the U.S. AI industry embraces practical AI (vs AGI or ASI), and the U.S. energy sector will be obliged to embrace AI at a breakneck speed.

Dubious assumptions, systemic biases, misperceptions, heroic guesses, and outright mistakes are distorting the views of key domestic and international stakeholders in both the energy and AI sectors:

- The ROI from general-purpose and frontier LLMs versus the payoff from inference and “mixture of experts” models, along with the impressive achievements of open-source models versus proprietary models, has introduced a high level of uncertainty into industry attempts to look forward. There is as much guesswork regarding which models will be monetizable and ultimately competitive as there was two years ago, perhaps even more.
- Global energy experts misunderstand, or fail to fully understand, the inherent scaling laws of foundation models, inference, and the emergent properties of advanced artificial intelligence more generally. The approach by the industry has been to forecast demand using fundamental analyses that look at everything ranging from announced data center projects plans, to real estate transactions, to forecasts issued by independent systems operators (ISOs). All this, in contrast to an approach that marries the fundamental analysis with the stated ambitions of the major AI companies in light of scaling laws. The forecasts of industry groups such as the IEA and EPRI, along with those of major investment banks, range widely. In general, however, such forecasts still come well below the demand that would be calculated from the declared ambitions of U.S.-based AI companies if observed from the perspective of scaling laws.
- AI industry leaders in the U.S., meanwhile, often underestimate the difficulty of retooling the nation's energy grid for the new era of AI, hyperscaling, and supercomputers, in part due to a blindness regarding regulatory and permitting headwinds, and in part due to poor insight into localized siting and supply chain problems. They also have a limited understanding of power markets,

which are esoteric and function in markedly different ways than other energy markets, such as oil. Some industry leaders are overly optimistic in believing they can readily get the electricity their companies will need on timelines that defy recent experience. Others realize the situation is bad in ways they are just beginning to understand. OpenAI founder and CEO Sam Altman observed in early 2024: “We still don’t appreciate the energy needs of this sector. There’s no way to get there without a breakthrough. We need [nuclear] fusion or we need like radically cheaper solar plus storage or something at massive scale.”¹¹ Such concerns have led to a sense of urgency that is arguably unprecedented in any peacetime economy of the United States. (NextEra Energy, a leading clean energy company and one of the world’s largest electrical utility holding companies, shared during its quarterly earnings call in June 2024, that a group of tech companies had approached it requesting 5 GW of capacity and sites that could accommodate such demand.¹² In its October call, it shocked markets by disclosing that it had secured framework agreements totaling 10.5 GW from two Fortune 50 companies (companies that are not from the technology industry, but wanted to get ahead of any bottlenecks anyway).¹³ Up to that point, companies with such large demand had not figured into renewable energy and storage forecasts for the balance of this decade. An agreement of more than 10 gigawatts is roughly equivalent to asking for enough additional electricity to power New York City and Los Angeles combined, or Chicago, Houston, and Dallas combined.)



- The “race” for artificial general intelligence is a misleading analogy. Artificial general intelligence and artificial superintelligence will not represent finish lines or end states, after which energy demand will plateau. They will likely usher in a further acceleration of domestic energy demand that may make current needs seem paltry.
- Both industry and government officials are too often proceeding with an apparently casual, face-value acceptance of Chinese government energy and

compute figures—which historically have been closely guarded, opaque, and difficult to benchmark against comparable U.S. data—as reflective of the reality on the ground in China.

- The Chinese government and Chinese firms have been conservative and reticent with technical claims and forecasts in recent history. U.S. players, perpetually in search of funding from venture and private equity funds, have a predilection for hyperbole. This has repeatedly caused a recurring series of misperceptions. Simply because the Chinese have had little to say amid a rapid industrial revolution, as in the case of the early phases of frontier AI model development, does not mean they will always have little to show.

The convergence of these errors and biases has led to an ill-considered view of the relative capabilities and trajectories of China and the U.S. AI industries. China has become a peer of the United States in many respects, and it is highly likely that, as of

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2025, China will take the lead along two key vectors: having more autonomy than the United States in controlling its AI destiny, and having strong inroads into the markets of nations of the global South.

Further, given the prevailing dynamics of U.S. industry (e.g., primarily closed, proprietary models rather than open source platforms; an emphasis on the AGI race rather practical, sector-specific applications), regulatory drag (long approval times for permitting, transmission, etc) and factor limits (volume and geographic location of rare earth minerals and compute power), U.S. industry may soon find that its ability to keep pace with the scaling, efficiency, and productivity of Chinese counterparts in critical sectors begins to lag, despite large pools of capital being committed.

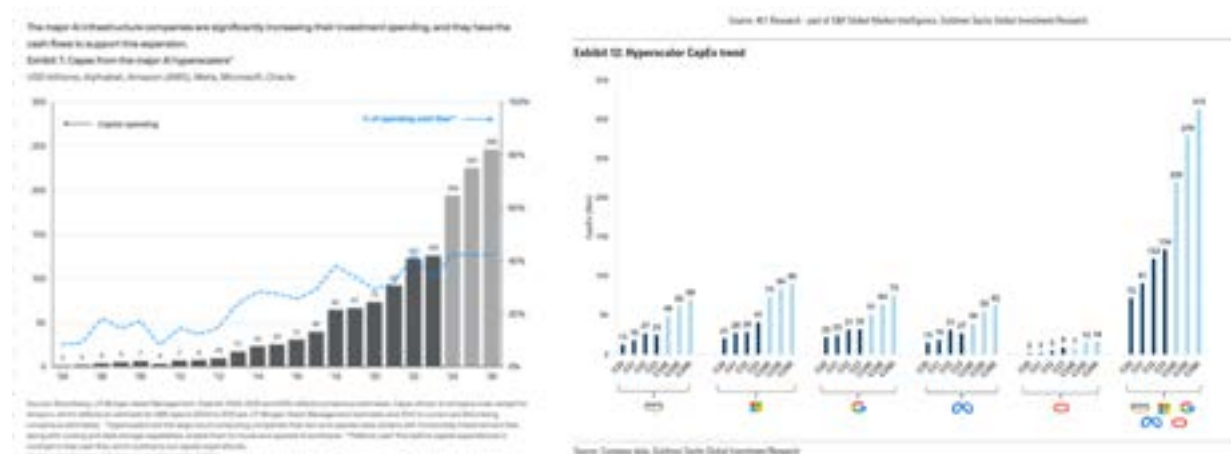
These factors, together with the interdependence of American and Chinese industries with respect to rare earths and chip foundries, point to the need for a comprehensive reevaluation of long-term competition and cooperation strategies by both the public and private sectors.

The Global Outlook

The quantitative, qualitative, and efficiency gains in the field of generative AI compute in recent years have outpaced other technological revolutions by a significant multiple. Industry analysts estimate that AI technology, at 4x annual

growth, is evolving twice as fast as that of mobile phone adoption, 1.5x that of solar energy capacity installation, and more than 3x of human genome sequencing.¹⁴

According to Crunchbase, in the span of a single year private venture and growth investment into AI companies was up 80 percent, growing from \$55.6bn in 2023 to more than \$100bn in 2024 (foundation companies are a subset of this number).¹⁵ Capital expenditures for AI infrastructure are equally remarkable in both absolute terms and as a percentage of leading companies' available financing.



Altogether, these investments will create a wave of productivity and follow-on capital that pushes far into the future at an exponential pace. Looking ahead, future investment for AI software and services alone has been estimated \$1.5 trillion in a lower range of scenarios in 2040 and to \$4.6 trillion in a higher range of scenarios, a CAGR of 17 to 25 percent.¹⁶ According to McKinsey, such investment, in turn, is estimated to generate as much as \$15.7 trillion in economy value for the global economy by 2030¹⁷ and \$23 trillion by 2040.¹⁸

Forecasting technological and market trends is notoriously difficult, not least because the number of variables from the research phase to the application phase is very high,

Venture and infrastructure investments in AI are of historic proportions right now, but there are numerous reasons to take forecasts with a note of caution.

such trends are inextricably intertwined with political, economic, and social dynamics, and deep expertise within sectors can lead to tunnel vision that misses material factors from outside the sector. Further, experts with access to

readily available data may be biased by personal or professional self-interest.

Companies, for example, may be overly optimistic in their projections in the hope of luring additional investment; academics may be overly optimistic to secure grants.

At the close of 2024, however, several bounding factors were coming into sharper relief:

- Marked differences in the performance of AI models, rapidly expanding in type and quantity, pointed toward a future shakeout as the weeding of models with less performance capability begins to shrink the number of players.
- The internet and other storehouses of publicly available information— notwithstanding their immense scale and ubiquity—will not yield an unending amount of usable text-based training data in massive chunks. Multimodal and spatial intelligence models, which require images, video, and audio, will require ever more compute and be significantly more expensive.
- There is a curse of geography, wherein available power generation sources faced capacity ceilings. Latency (caused by the need to access compute and power from distant sites) is likely to become more of pressing issue this year and next.

At the same time, 2024 saw “a blizzard” of major steps forward that indicate both staying power and an accelerating maturity in AI technology: progress in energy efficiency of semiconductors and data center cooling; LLM progress in accomplishing agentic workflows and application in B2C devices and services; significant gains in inference engines and increasingly realistic manipulation of ever-longer form video content, which is extremely compute-intense; decreasing prices due to a combination of the rapid pace of innovation in algorithmic efficiency and increased competition; continued development of science-related apps with clear use cases such as AlphaDev, GnoME, and GenCast; major breakthroughs against performance tests; strides forward in model efficiency; and new or upgraded strategic partnerships among key players.¹⁹

Repercussions of Perpetually Scaling Frontier Models

Consider the matrix of variables a financial industry analyst looks out upon in trying to come up with a credible macro forecast. From the outset, there is an apples-to-oranges challenge. Training models have massive compute, power, and externally sourced data requirements, while inference models are typically characterized by high-bandwidth connections, located in an ideal world as close as possible to reduce latency for end-user applications and leverage additional, often proprietary, datasets.

Even if such tradeoffs can be squared, one must then put such assumptions through the next sieve of possibilities: the proliferating array of data centers. There are hyperscale data centers, wholesale data centers, co-located retail centers, and corporate-level data centers, each of which has evolving capabilities (e.g., average power usage effectiveness (PUE)²⁰). Moreover, customers have the ability to choose between, say, seeking efficiency gains in legacy equipment, investing in new bespoke capacity, or buying compute from vendors. Dedicated AI data centers are designed and built for training workloads, and they operate in fundamentally different ways than those designed for more traditional workloads.

Data centers are bespoke in many ways. Small-scale data centers cater to small businesses or government agencies; “edge” data centers are strategically located near the outer edges of existing networks to serve geographically distant clients; enterprise data centers are owned and operated by single large companies for exclusive use; and co-location centers incorporate several businesses that may rent space to house servers and share energy and cooling infrastructure.²¹

Further, models can be tailored for cost, efficiency and optimization by making tradeoffs between speed, accuracy, precision, and size of data sets, by introducing innovations in algorithms and hardware, and by investing in higher-quality data sets. An increasing number of functions may move from data centers to on-device platforms, such as mobile phones.

At the heart of the bewilderment over AI’s overall trajectory is the fact that the workings of large language models are inherently difficult to comprehend, much less predict, for all but a few specialized engineers and industry experts. Multimodality simply increases complexity, and complexity increases confusion.

The challenge starts with data. It’s not only that more data requires more power; it is that the edges of the data universe are more complex and progressively harder to train than the core. Alphanumeric data sets are easier to train than images, which in turn are easier to train than audio, which in turn is easier to train than video. The curve of problem-solving intensity arcs upward, not down.

More and more compute, of course, requires ever more power. How much more? Between 2017 and 2021, even before the generative AI explosion of demand, electricity used by Meta, Amazon, Microsoft, and Google alone—the main providers of commercially available cloud computing and digital services—more than doubled.²²

From 2024 to 2030, the amount of electricity demanded by data centers in the United States could more than double again.

Rapid evolution of datasets, training techniques, and model architecture complicates estimates and comparisons. Scaling laws are not universal across models and may be poorly understood within models.²³ The performance of inference models using LLMs can vary significantly depending upon the task given and deployment environment. Those who design and train such models have considerable leeway to make tradeoffs between factors such as efficiency, latency, and performance optimization. This is just a partial list of challenges facing forecasters.

Nonetheless, gross “order of magnitude” (OOM) changes in compute (one OOM is defined as a 10x increase in compute requirements)²⁴ have been calculated with considerable plausibility in some corners of the AI industry in recent months and they point compellingly toward OOM changes ahead.^{19 20}

GPT-2 to GPT-4 took us from ~preschooler to ~smart high-schooler abilities in 4 years. Tracing trendlines in compute (~0.5 orders of magnitude or OOMs/year), algorithmic efficiencies (~0.5 OOMs/year), and “unhobbling” gains (from chatbot to agent), we should expect another preschooler-to-high-schooler-sized qualitative jump by 2027—a jump that takes us from high schooler to a smart AI engineer²⁵

Such a jump will come at a significant cost. OpenAI’s GPT-4 was trained on an estimated 10,000 H100 equivalent cluster and consumed around 10 MW of power. Last year alone, that feat was exceeded by 10x: Elon Musk’s xAI, required a cluster of some 100,000 H100s and consumed 100 MW of power. Assuming the scaling laws proposed by Epoch AI²⁶ and those calculated in the widely circulated “Situational Awareness” paper²⁷ are plausible, the next 24 months could usher in a 5 GW cluster that costs tens of billions of dollars to train.

More broadly speaking, the tight race amongst frontier model companies in the U.S. alone means that, by 2027, one could see three to five players requiring 5 GW data centers standalone each (to avoid multi-site training, which is more costly and less efficient). To put this into perspective, only one power plant in the U.S. today generates more than 5 GW—the hydroelectric facility at the Grand Coulee Dam, which took more than a decade to build from 1933-42.²⁸

Year	OOMs	H100s-equivalent	Cost	Power	Power reference class
2022	~GPT-4 cluster	~10k	~\$500M	~10 MW	~10,000 average homes
~2024	+1 OOM	~100k	\$billions	~100MW	~100,000 homes
~2026	+2 OOMs	~1M	\$10s of billions	~1 GW	The Hoover Dam, or a large nuclear reactor
~2028	+3 OOMs	~10M	\$100s of billions	~10 GW	A small/medium US state
~2030	+4 OOMs	~100M	\$1T+	~100GW	>20% of US electricity production

The Power Conundrum

The past several years have witnessed some of the most extraordinary leaps in predictive artificial intelligence in the history of humanity. The irony is that analysts trying to predict the future of the artificial intelligence sector itself—and, in particular, its power needs—seem unable to converge on anything like a consensus view of its future.

Even a cursory survey of major industry reports finds a remarkably wide distribution of estimates. The “lowest of the low” forecasts for the global power demand of data centers to be around 68 GW by 2030, while the highest of the high is close to 300 GW. Both these, derived from a current level of global demand estimated to be around 55-60 GW.²⁹

ESTIMATES OF GLOBAL POWER DEMAND BY DATA CENTERS					
	GS	JPM*	IEA	McKinsey	EPRI*
2022			52.5 GW		
2023				60 GW	53 GW
2024	55 GW				
2025					
2026			91.3 GW (Base)		
2027	84 GW (Base)	103 GW			
2028		117 GW			
2029					
2030	118 GW (Base)			219 (Mid) - 298 GW (High)	75 (Mid) - 103 GW (High)
<i>Note: Author survey of research by The Goldman Sachs Group Inc., JP Morgan, International Energy Agency, McKinsey & Company, and Electric Power Research Institute Inc. *EPRI global estimate is inferred based on U.S. proportion of global demand. JPM calculation uses JPM supply forecasts with Goldman Sachs capacity utilization rate forecasts.</i>					

A dissection of the numerous forecasts is beyond the scope of this paper. There are plausible historical reasons to think some forecasts may be too low, such as the fact that technological breakthroughs with application across every industrial sector (e.g., fire, steam, electricity) rarely come along.

Another reason why current energy industry forecasts of power demand may be low—and inscrutable—is that a considerable amount of AI spending is intentionally hidden in government “black” budgets for intelligence-gathering, counterintelligence, cybersecurity, digital warfare, and the like around the world.

According to recent estimates, the U.S. Department of Defense had some 657 artificial intelligence programs yet a comprehensive review of hundreds of publicly accessible DoD contracts shows a spend of around ~\$4.5 billion per annum on AI. At face value, this would imply the most militarily powerful nation on earth is spending an average of ~\$6.8 million per contract—a figure that does not pass the test of common sense.³⁰ The U.S. government—along with the governments of China, Russia, European nations, India, and others—are highly likely to be spending multiples of what is publicly reported. Furthermore, the associated data center use would not be captured in publicly reported numbers.

At the same time, the fundamental supply forecasts of both AI experts and energy experts—those that attempt to assess on-the-ground factors and build models from the bottom up—seem to be overlooking some hard realities.

- AI industry forecasts of power needs have commonly been based on direct needs for the amount of electricity needed to support training compute, ignoring the amount of ancillary energy required for cooling, other IT equipment, and facility operations, etc. These, in some instances, can consume almost as much as energy as the compute itself.
- Energy planners and forecasters have tended to focus on the gross amounts of power required at the sector or national level, while glossing over localized constraining factors, such as siting, grid connection times, labor shortages, permitting, natural disasters, and port strikes.
- Shortages of key components are already beginning to bite. There are generator backlogs, waiting lists for natural gas turbines, and longer lead times for crucial data center components such as switchgears, power distribution equipment, and chillers.³¹
- The lead times for power transformers alone have increased from roughly one year as of 2021 to more than two years as of 2024. Large transformers, both substation power and generator step-up transformers, have lead times ranging from 80 to 210 weeks, according to global industry consultant Wood Mackenzie.³²
- An astounding 93 percent of data center respondents already report some level of supply chain constraints, and 21 percent are experiencing outages.³³ Training workloads need uninterruptible power supply and investment in base load power will be critical.

DATA CENTER SUPPLY CHAIN CONSTRAINTS AS MEASURED BY EXTENDED LEAD TIMES	
REPORTED LEAD TIMES	COMPONENTS
0.5 - 1 Year	Networking hardware; batteries; uninterruptable power supplies (UPS); medium-voltage switchgears;
1 - 1.5 Years	Chillers; IT servers; generator enclosures; UPS containers; low-voltage switchgears.
1.5 - 4 Years	Large sub-station power transformers; large generator step-up transformers; grid connections; busways; piping; copper feeders
2- 3 Years	Data Center Generators
4 Years	Large Natural Gas turbines

Multiple Sources³⁴

The power industry writ large, outside of markets such as India, China, and the Asia-Pacific region, has had anemic growth rates for decades. In a short period of time, it is being asked to plan for the speed and scale of the AI industry. Goldman Sachs, for example, estimates that, relative to 2023, global data center power demand will grow by 165% by 2030, and there will be 122 GW of data center capacity online, which would require a CAGR of 14 percent.³⁵ In contrast, JP Morgan forecasts that there will be 122 GW of capacity by 2028. Other estimates are discussed later in this report.

If, based on the current trajectories of project development, much of the additional capacity in energy generation will come via clean energy sources, there is good news and bad news, according to IEA.

The good news: with respect to current projects alone, an estimated \$180 billion will be invested per year between 2025 and 2030, and roughly one third of that figure has been committed.

The bad news: this figure, together with the normal pace of new projects announced each year, is significantly out of proportion to demand forecasts. The numbers simply don't add up to a sustainable trajectory, and "some decline in the rate of investments likely to occur."³⁶

Overview of U.S. Position

Given that the United States has historically had a commanding research and development position in artificial intelligence, and this has led to record-setting levels of investment in recent years, the preponderance of which has gone to U.S. companies.

According to Pitchbook data, a record of more than \$40 billion was invested into foundation models and other core AI startups this year, bringing the total raised since the start of 2023 to \$60.2 billion.³⁷ Those dollars are increasingly concentrated in a small cadre of leading companies. Deals for OpenAI, Anthropic, and xAI in 2024 nearly matched the total invested the previous year.

North America-headquartered generative AI software vendors alone represented over \$25 billion in funding in 2023, or 88% of global funding tracked for the year, the implication is that US industry will retain its dominance as the industry scales to an estimated \$52 billion by 2028.³⁸

In gross terms, the United States has some 1,300 GW of generation capacity, of which roughly 44 percent comes from natural gas and 16 percent comes from coal. Wind, nuclear, hydro, and solar combined account for about one-third of capacity.³⁹

Progress in adding capacity, however, is almost literally “two steps forward, one step back”. For 2023, the most recent data available, the U.S. energy sector added more than twice as much capacity as it retired (36 gigawatts vs 16 gigawatts), and it

The U.S. has a commanding lead in semiconductor manufacturing, number of AI companies, influential research, and world-beating ‘unicorns’. But those are not the only metrics that matter.

cancelled as much in capacity additions (36 gigawatts) as it added. Solar accounts for the plurality of capacity added (48%) and a plurality (47%) of cancellations by megawatt.⁴⁰ From 2016 to 2023, the net addition of capacity (additions less retirements) was 109 GW while

cancellations totaled a whopping 340 GW, making for 3x more in cancelled GW than added. That is a lot of churn for any industry, especially one so critical to the nation’s overall economic trajectory.



SOURCE⁴¹

The United States has an AI talent pool unmatched in scale and diversity that attracts and retains new talent from academia, other industrial sectors, and other nations. For example, while China has for years outpaced the U.S. in the volume of its AI publications across the board, in the sub-sector of generative AI the two nations are essentially tied, but the US leads markedly in arguably the most important category, *top-cited* generative AI research.⁴² It has a mature industrial ecosystem that cross-pollinates ideas, model advances, talent, market signals, and technology among companies, platforms, infrastructure providers, academia, and government agencies.

As of the start of this decade, the U.S. could boast some 2,000 AI companies, and countless others that employed AI technologies, more than double its closest competitor, and a preponderance of AI “unicorns,” privately-owned startup companies that reach a valuation of \$1 billion without being listed on the stock market.⁴³ For the decade from 2014-2024, the U.S. is estimated to have ballooned to nearly 5 times as many AI companies as the Chinese mainland.⁴⁴ These companies are at the forefront of refining and commercializing AI technology for global markets.

Table 1: Overview of AI companies, investments, and investment value 2014–2024¹

Metrics (Absolute Number or Value)	China (Mainland)	United States
Number of companies	1,944	9,500
Number of incoming investments	8,194	59,534
Estimated total value of incoming investments (millions)	\$85,650	\$605,416

SOURCE: ITIF

A comprehensive transnational study conducted by the OECD found that U.S.-based AI venture startups held a pronounced lead in the number of deals and dollars invested, so much so that they represented more than half of investment worldwide, i.e., more than all other competitors combined.⁴⁵ This study was conducted before the release of ChatGPT in November 2022, which further turbocharged American investment.

The U.S. has been the pacesetter in key frontier areas of AI such as semiconductor design, cloud computing, and quantum information processing. In particular, the U.S. garners roughly 40 percent of the economic value of the global semiconductor supply chain.⁴⁶ So-called “hyperscaler” cloud-service providers based in the US—such as Amazon Web Services, Google Cloud, and Microsoft Azure—are well-positioned to take advantage of the robust global growth in demand for cloud services.⁴⁷

Hardware advantages, in part, have contributed to America’s pronounced leadership position in machine learning and foundation models, of which US entities have produced roughly four times as many as their Chinese counterparts.⁴⁸

Public policy matters. In recent years, the U.S. executive and legislative branches—motivated in part by China’s rapid advances in AI—took considerable steps to ensure America’s leading role in AI research, development, and application:

- Three executive orders across the first Trump administration and the Biden Administration were issued to promote AI R&D, establish broad common principles for AI design, development and application of AI in the federal government, and ensure a government-wide coordination of federal agencies.⁴⁹
- Congress passed legislation, subsequently signed into law by the president, in December 2020 (the AI in Government Act) and December 2022 (the Advancing American AI Act, and the CHIPS and Science Act) to promote American AI.⁵⁰ The funding and support via the CHIPS and Science Act are expected to significantly boost US semiconductor manufacturing capacity in the coming years, especially in advanced chips (less than 10nm).
- A bipartisan working group of the U.S. Senate compiled and issued a wide-ranging assessment intended to serve as a policy roadmap for the coming years.⁵¹
- Just before exiting office in January 2025, President Joe Biden issued an unprecedented executive order allowing the Department of Defense and Department of Energy to make government land available for the construction

of new data center campuses and transmission infrastructure via expedited leases.⁵²

A McKinsey study estimates that, altogether, this will represent an additional \$280 billion in federal funding over the coming years to spur U.S. semiconductor capacity, create additional high-tech hubs, and enhance America's STEM workforce.⁵³

America's track record of relative macroeconomic and political stability as a substrate for its AI industry represents an important intangible asset: a majority of executives in advanced technology companies report that geopolitical stability and supply chain disruptions are negative risks in the near- to mid-term future.⁵⁴

In addition, the U.S. has been forward-leaning in using export controls and its geopolitical influence to slow or thwart Chinese technological inroads into an array of developed (e.g., Canada, Europe, South Korea, UAE) and developing nations (India, Mexico, Ghana),^{55 56} and to shape governance standards worldwide.

Overview of China's Position

China's advantages and disadvantages in AI research and development are considerable and, in many respects, the opposite of those that characterize the US AI sector.

The U.S. brings to the AI global evolution a sprawling, decentralized market-based approach centered on innovation. This has been aided, and in some cases bedevilled, by a loosely organized and slowly evolving regulatory environment.

China, meanwhile, has employed a much more centralized, state-driven strategy that has rapidly directed massive subsidies to key industry sub-sectors and state-owned enterprises where AI applications can more quickly be applied to use cases and-or monetized. Alexandr Wang, the co-founder and CEO of Scale AI, a prominent data annotation platform for machine learning models, estimated in early 2025 that China is outspending America by 10 times on AI implementation and adoption.⁵⁷

In a field where economies of scale are a major factor, China bests the United States handily in key metrics. China is the largest producer and consumer of energy in the world. As such, its decisions regarding choice of technology, subsidies, and industries to prioritize have enormous consequences for suppliers and consumers alike. Between 2019 and 2024, China accounted for an estimated 40% of the global renewable energy

capacity expansion, and almost half of the distributed photovoltaic growth, overtaking the EU as the world leader in the latter category. Unlike the U.S., more than 60 percent of China's energy generation comes from coal plants.⁵⁸

Of note, China is the cheapest manufacturer of clean energy technologies, even if one takes government subsidies out of the equation, according to the International Energy Agency.⁵⁹ According to a recent IEA report, "Compared with China, it costs up to 40% more on average to produce solar PV modules, wind turbines and battery technologies in the United States, up to 45% more in the European Union, and up to 25% more in India."⁶⁰ This confers a major advantage, as China can more rapidly expand its clean tech economy than the U.S. at a time when both nations are under pressure to both address climate change and accommodate increased demand from the artificial intelligence sector.

While China has a high import burden for the purchase of fossil fuels, notably natural gas, the same report found that China's share of global manufacturing for the six major clean technologies (solar PV, wind, electric vehicles, batteries, electrolyzers, and heat pumps), as measured by dollar value, is around 70% and its investment in such technologies jumped from \$160 billion to \$235 billion from 2022 to 2023 alone.

Far from a steady neck-and-neck competition, the race between the United States and China in battery and solar PV manufacturing has seen China expand its lead in recent years via major investments in anodes, cathodes, polysilicon, and wafers.⁶¹

Figure 1.6 Net manufacturing capacity additions for selected clean energy technologies by country/region, 2020-2023

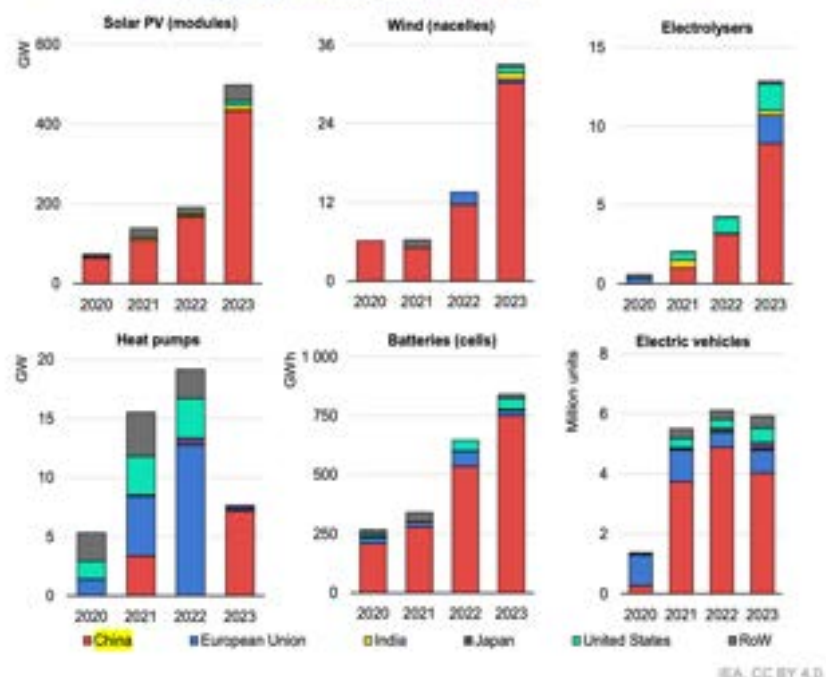
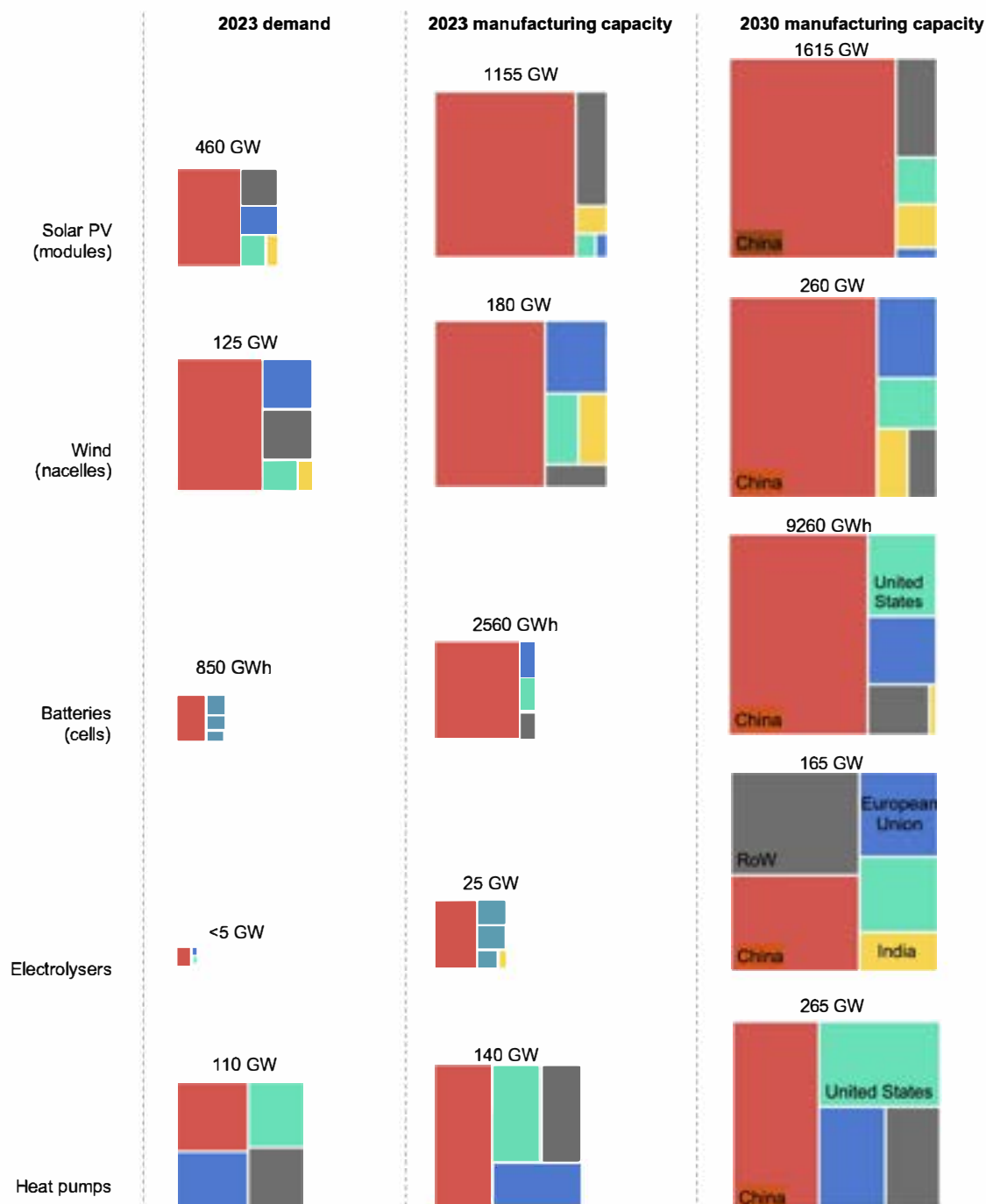


Figure 1.8 Global demand and manufacturing capacity by country/region for selected clean energy technologies based on announced projects, 2023-2030



IEA. CC BY 4.0.

Notes: RoW = Rest of World. Capacity at year-end. 2030 manufacturing capacity includes existing and planned capacity based on project announcements. The size of each square is proportional to demand/capacity, with each row scaled relative to 2030 manufacturing capacity.

Sources: IEA analysis based on S&P Global (2024); WindEurope (2023); BNEF (2024a); GWEC (2023); Wood Mackenzie (2024); InfoLink (2024); SPV Market Research (2024); BMI (2024); EV Volumes (2024); BNEF (2024b); BNEF (2024c); Oxford Economics Limited (2024b); CEPII (2024); as well as announcements by manufacturers and personal communications, gathered by the IEA.

China's 1.4 billion citizens—a combined population, test bed for innovations, and diverse market—is more than four times that of the United States. This confers an obvious advantage with respect to the amount of training data available, especially insofar as China has fewer privacy protections for the data of its citizenry.

China has been graduating roughly seven times as many engineering students as the U.S. academic system (1.38 million vs. 197,000).⁶² As of 2025, enrollment patterns indicate that China will be churning out roughly double the number of STEM PhD's as the United States (77,000 versus 40,000), and some 45% of China's STEM PhD's will hail from its elite universities.

Put another way, China is annually producing almost as many elite STEM experts as the U.S. is producing from all its universities and colleges combined. This has led some experts to conclude that China's STEM talent pool exceeds domestic demand, even as the U.S. is trying to close a widely reported STEM gap between supply and demand.⁶³

The race for patent filings is an obviously crude metric, as it does not capture the qualitative aspects of the patents being filed. Nonetheless, it does provide helpful snapshot of how China's combination of national champions and academic institutions has outpaced the U.S. market-led approach.⁶⁴

With respect to factors of production, a small number of rare earth elements (REE) are critical to artificial intelligence technology: neodymium, erbium, terbium, dysprosium, samarium, cerium, and praseodymium. U.S. officials recently acknowledged that China has “cornered the market cornered the market for processing and refining of key critical minerals, leaving the U.S. and our allies and partners vulnerable to supply chain shocks and undermining economic and national security.”⁶⁵

Numbers support this conclusion. China holds an estimated 60 percent of global REE production⁶⁶, 85-90 percent of REE processing capacity⁶⁷, and an estimated 99 percent of heavy rare earth separation capacity in dysprosium⁶⁸.

The global level of proven reserves of rare earth elements is estimated to be roughly 130 metric tons (MT), of which China has 44 MT (34%), Russia has 10 MT (8%), and the US has 1.8 MT (>2%)⁶⁹. Africa, meanwhile, where China has established a stronghold in the mining sector, is coming on strong. Mines in countries such as Tanzania, Angola,

Malawi and South Africa are expected to start production by 2029 and contribute 9% of global supply, according to Benchmark Mineral Intelligence.⁷⁰

Further, China has demonstrated a willingness to use a combination of stockpiling, quotas and, in some cases, tit-for-tat trade bans in order to further its market dominance.⁷¹ Within hours of a White House announcement in December 2024 that it would be tightening bans on advanced semiconductor exports, China announced that it would begin banning rare earth elements such as gallium, germanium, antimony and so-called superhard materials.⁷²

While the United States federal government has a pro forma regulatory neutrality toward companies, China began as early as 2017 designating “national champions” in

China’s early bets on national AI champions, energy infrastructure, and open-source models are paying off. And its chokehold on rare earth elements is formidable.

artificial intelligence, with some success. In 2017, China designated Baidu, Alibaba, TenCent, and iFlyTek as the core of its AI group singled out for favorable treatment. By the end of the 2010s decade, it added 10 additional

national champions, including SenseTime, Huawei Technologies, Hikvision Digital Technology, Xiaomi Corp, JD.com, Qihoo 360 Technology, Megvii, and Yitu⁷³. China has repeatedly set ambitious public targets for its AI progress as a nation, in 2020, 2025, and 2030, raising the domestic political stakes for measuring progress.

In 2021, the Chinese government began an initiative to explicitly encourage these companies to pursue open-source AI platforms to encourage cross-company, cross-agency, and public-private collaboration that would improve the nation’s overall AI trajectory. The plan’s intent was “to support the development of innovative consortia such as digital technology open source communities...encourage enterprises to open up software source code, hardware designs, and application services. This coincided with a wave of government action in open-source software, including support for Gitee, an alternative to the popular developer collaboration platform GitHub.”⁷⁴

This bet is beginning to yield dividends, as Alibaba’s Qwen 2.5 and Zhipu AI’s ChatGLM3, among others, have outperformed some U.S. counterparts and gained recognition for their impressive capabilities.⁷⁵

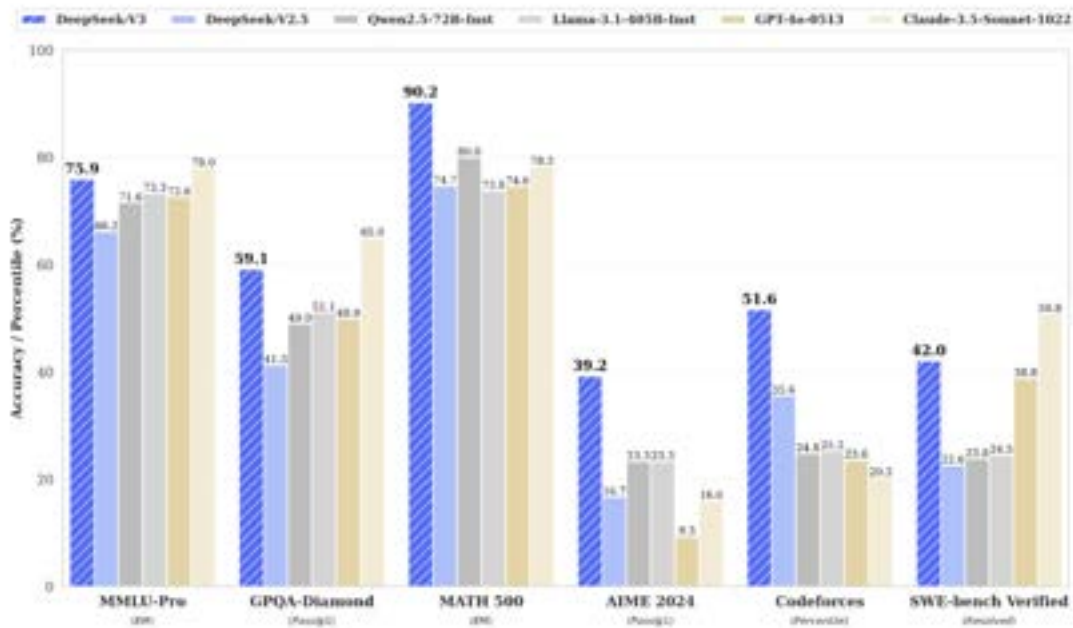
Open platforms have notched qualitative as well as quantitative gains. As the new year turned, DeepSeek released an DeepSeek-V3 followed by DeepSeek R1, an open-source “mixture of experts model” that bested the performance of leading U.S.-based

models such as Llama-3.1, ChatGPT-4o, and Claude-3.5 by, among other attributes, employing 671B total parameters and pre-training on 14.8 trillion tokens.⁷⁶

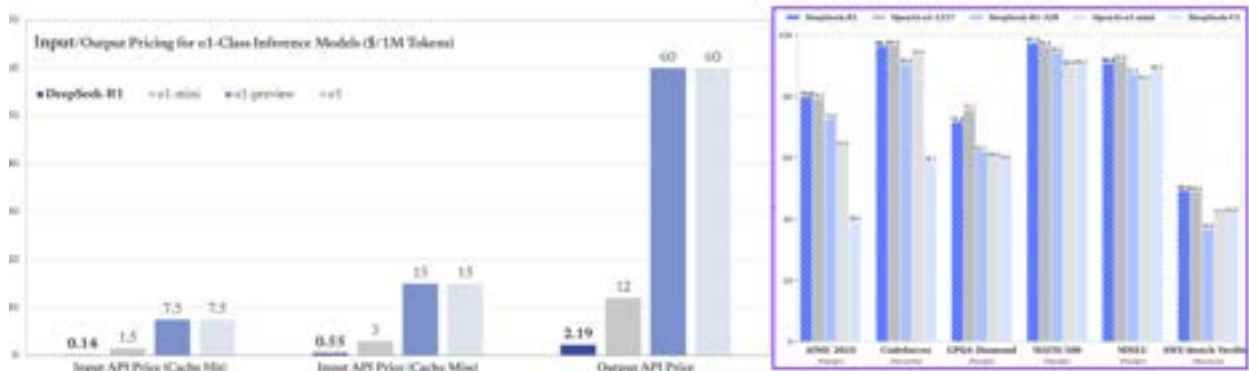
Of note, the model required only 2.788M H800 GPU hours for its full training,⁷⁷ and it cost a fraction of what U.S.-based competitors cost to train (less than 2 percent of what it cost to train Meta's Llama 3.⁷⁸ Deepseek-V3 is being heralded in some quarters as a "rising star" likely to have a staying power in leaderboards that other models have struggled to achieve, in part because of its open-model approach.^{79 80}

Arnaud Bertrand, a knowledgeable industry observer and co-founder and CEO of Me & Qi, posted recently about DeepSeek R1: "It's essentially as if someone had released a mobile [phone] on par with the iPhone but was selling it for \$30 instead of \$1,000. It's this dramatic. What's more, they're releasing it open-source so you even have the option—which OpenAI doesn't offer—of not using their API at all and running their model 'free' yourself. If you're an OpenAI customer today, you're obviously going to start asking yourself some questions, like 'Wait, why exactly should I be paying 30x more?' This is pretty transformational stuff. It fundamentally challenges the economics of the market."⁸¹

China, in general, continues to make larger contributions to open-source AI than U.S. industry. Alibaba Cloud alone has released more than 100 open-source models in more than two dozen languages, and these models collectively have been downloaded more than 40 million times from platforms such as Hugging Face and Modelscope.⁸² In specific domains such as processing of visual data, Alibaba Cloud has also demonstrated impressive gains, such as the ability to comprehend videos more than 20 minutes long, process prompts more efficiently, and include text in both Chinese and English.⁸³



SOURCE: DEEPSEEK-V3-BASEPAPER



SOURCE: Arnaud Bertrand

According to Hugging Face, China's AI model Qwen became the most downloaded AI model in 2024. In one prominently reported instance, a U.S.-based team built a high-performing model on top of Chinese open-source building blocks. Why? Chinese building blocks were preferred.⁸⁴

Explore the community's favourite models, ranging from tasks, likes to downloads, since 2022. What we're seeing: smol models are on the rise. [👁️++](#)

This treemap visualizes the distribution of AI model downloads across various categories and time periods. The data is organized into a hierarchical structure, with the largest categories being 'Natural Language Processing' (NLP) and 'Computer Vision'. The 'NLP' category is further divided into 'Language Models' and 'Speech Recognition'. The 'Computer Vision' category is divided into 'Image Classification' and 'Image Segmentation'. The 'Language Models' category is the largest, accounting for 51.2% of the total downloads, and is further divided into 'Generative Models' (28.1%) and 'Language Understanding' (23.1%). The 'Generative Models' category is the largest within NLP, accounting for 28.1% of the total downloads, and is further divided into 'Text-to-Text' (15.2%) and 'Text-to-Speech' (12.9%). The 'Text-to-Text' category is the largest within Generative Models, accounting for 15.2% of the total downloads, and is further divided into 'Qwen2.5-1.5B-Instruct' (15.2%) and 'GPT-4o' (2.94%). The 'Text-to-Speech' category is the largest within Generative Models, accounting for 12.9% of the total downloads, and is further divided into 'TTS-English' (3.02%) and 'TTS-French' (2.19%). The 'Language Understanding' category is the largest within NLP, accounting for 23.1% of the total downloads, and is further divided into 'Intent Classification' (2.06%) and 'Sentiment Analysis' (2.00%). The 'Image Classification' category is the largest within Computer Vision, accounting for 17.5% of the total downloads, and is further divided into 'ImageNet' (1.75%) and 'ImageNet-V2' (1.48%). The 'Image Segmentation' category is the largest within Computer Vision, accounting for 1.75% of the total downloads, and is further divided into 'PASCAL3D+' (1.75%) and 'PASCAL3D+' (1.48%). The 'Speech Recognition' category is the largest within NLP, accounting for 1.75% of the total downloads, and is further divided into 'ASR-English' (1.75%) and 'ASR-French' (1.48%). The 'Image Classification' category is the largest within Computer Vision, accounting for 1.75% of the total downloads, and is further divided into 'ImageNet' (1.75%) and 'ImageNet-V2' (1.48%). The 'Image Segmentation' category is the largest within Computer Vision, accounting for 1.75% of the total downloads, and is further divided into 'PASCAL3D+' (1.75%) and 'PASCAL3D+' (1.48%). The 'Speech Recognition' category is the largest within NLP, accounting for 1.75% of the total downloads, and is further divided into 'ASR-English' (1.75%) and 'ASR-French' (1.48%).

Category	Sub-Category	Model	Percentage
Natural Language Processing (51.2%)	Language Models (28.1%)	Qwen2.5-1.5B-Instruct	15.2%
		GPT-4o	2.94%
		TTS-English	3.02%
		TTS-French	2.19%
		ASR-English	1.75%
	Language Understanding (23.1%)	Intent Classification	2.06%
		Sentiment Analysis	2.00%
		ImageNet	1.75%
		ImageNet-V2	1.48%
		PASCAL3D+	1.75%
Image Classification (17.5%)	ImageNet	1.75%	
	ImageNet-V2	1.48%	
	PASCAL3D+	1.75%	
	PASCAL3D+	1.48%	
	PASCAL3D+	1.75%	
Image Segmentation (1.75%)	PASCAL3D+	1.75%	
	PASCAL3D+	1.48%	
	PASCAL3D+	1.75%	
	PASCAL3D+	1.48%	
	PASCAL3D+	1.75%	
Speech Recognition (1.75%)	ASR-English	1.75%	
	ASR-French	1.48%	
	ASR-English	1.75%	
	ASR-French	1.48%	
	ASR-English	1.75%	

Created by [AI Works](#) and [Hustle Free](#)

More generally, independent analysts have estimated that, among the top 30 or so LLMs, some two-thirds of the models are of Chinese origin.⁸⁵

Figure 2: Scores of Chinese and U.S. LLMs on Chinese-language understanding based on SuperCLUE benchmark⁷⁷

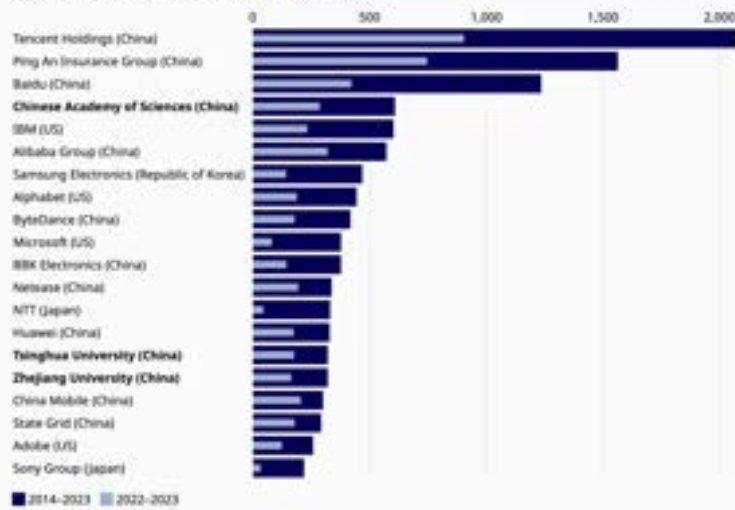
April 2024		June 2024	
Model	Score	Model	Score
GPT-4-Turbo-0125	79	GPT-4o	81
GPT-4-Turbo-0409	77	Claude-3.5-Sonnet-200K	77
GPT-4	75	Qwen2-72B-Instruct	77
Claude3-Opus	74	DeepSeek-V2	76
Baichuan3	73	GLM-4-0520	76
GLM-4	73	SenseChat5.0	76
通义千问2.1	72	AndreyGPT	75
腾讯混元-pro	72	GPT-4-Turbo-0409	75
文心一言4.0	72	GPT-4	73
Moonshot(Kimi)	70	Baichuan4	72
从智大模型V1.5	70	Doubao-pro-32K-0615	72
MiniMax-abi6.1	70	Moonshot(Kim0)	72
山海大模型	70	山海大模型4.0	72
讯飞星火V3.5	69	360gpt2-pro	72
Llama-3-70B-Instruct	69	MiniMax-abi6.5	71
字节跳动step-1-32K	69	通义千问2.5	71
qwen-1.5-72B-chat	68	文心一言4.0	69
讯飞星火	67	Llama-3-70B-Instruct	69
360gpt-pro	67	Gemini-1.5-Pro	68
GPT3.5-Turbo-0125	67	字节跳动step-1-32K	68
Gemini-Pro	64	讯飞星火V4.0	67
qwen-1.5-14B-chat	64	GPT-3.5-Turbo-0125	64
Llama-3-8B-Instruct	57	Yi-1.5-34B-Chat-16K	63
XVERSE-13B-L	56	qwen2-7B-instruct	62
qwen-1.5-7B-Chat	55	XVERSE-65B-2-32K	60
Llama-3-70B-Instruct	54	qwen1.5-32B-chat	57
Baichuan2-13B-Chat-v2	53	Llama-3-8B-Instruct	54
ChatGLM3-6B	51	Baichuan2-13B-Chat-v2	53
Gemma-7B	44	Yi-1.5-6B-Chat	53
Chinese-Alpaca2-13B	44	Phi-3-mini-128K-Instruct	40
Llama2-13B-Chat	41	Gemma-7B	38
Llama2-7B-Chat	40	qwen2-1.5B-instruct	38
		Llama-3-13B-chat	35

SOURCE:ITIF

While China has far fewer AI companies than the U.S., America’s position in the actual development and use of AI intellectual property—as measured by patent activity—is well behind both Chinese companies and Chinese academic institutions. Indeed, the World Intellectual Property Organization found that, from 2017 to 2024, China published more patents in AI each than all other countries combined.⁸⁶

Although corporations dominate as the largest patent owners, the Chinese Academy of Sciences, a research organization is fourth in the list, above IBM (research organizations shown in bold).

Figure 12 Top patent owners in GenAI, 2014–2023



Note: Published patent families in GenAI.
Source: WIPO, based on patent data from EcomSight/IFI Claims, April 2024. • Get the data

SOURCE: ITIF

There is a dominance of Chinese universities/research organizations along with a few from the US, Republic of Korea, Japan, and Switzerland in the top 20 patent owners among universities/research organizations.

Figure 13 Top patenting research organizations in GenAI, 2014–2023



Note: Published patent families in GenAI.
Source: WIPO, based on patent data from EcomSight/IFI Claims, April 2024. • Get the data

SOURCE: ITIF

If one is in search of litmus-test statistics to watch among the sea of data on the U.S.-China AI rivalry, two candidates may be data center racks and supercomputer exa-FLOPs (floating-point operations per second).

While the U.S. boasts a larger number of data centers, China may have closed the gap in data center racks. (Data centers are facilities, the chassis of artificial intelligence; data center racks are the engines inside). Chinese entities do not disclose as many AI statistics as U.S. counterparts. However, China clearly has many efficient, later-generation semiconductor chips (“leapfrogging” legacy U.S. capacity) that in the aggregate are consuming more energy than those in the U.S. This may mean that it has fewer data “centers” but more muscle in terms of data center racks and square footage.⁸⁷

There are good reasons to believe the estimates of gross compute available to China—as measured via the reported numbers of Chinese data centers—are low. As early as four years before the explosion of interest in generative AI, the Chinese government

China may lag behind the U.S. in the number of ‘data centers’ but that should not be confused with its de facto access to gross compute.

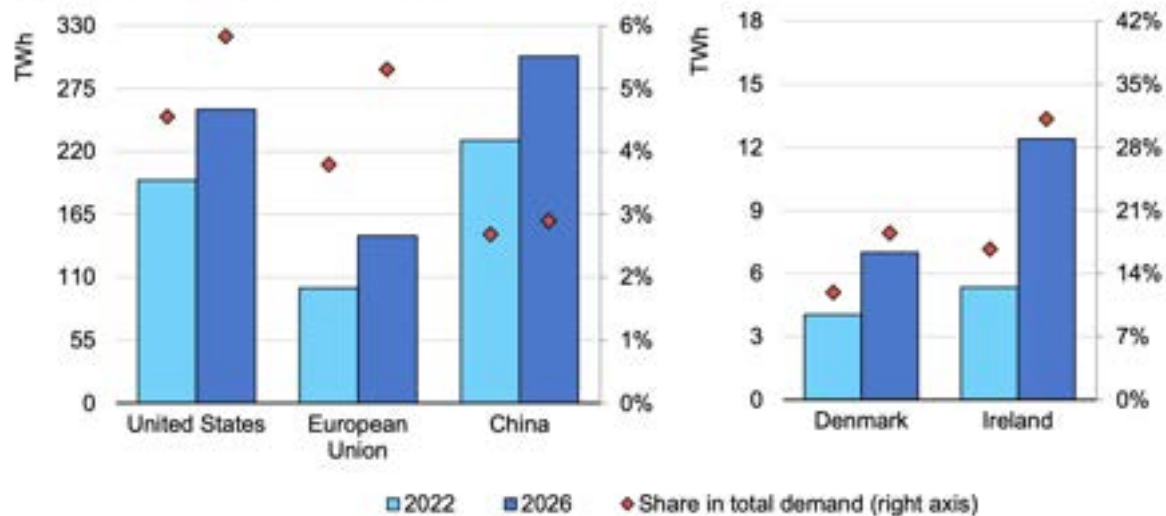
was a global leader in mainstreaming advanced AI technology in massive social applications: facial recognition technology for both real-time and predictive policing;⁸⁸ personalized education and tutoring through platforms such as Squirrel AI;⁸⁹ and the so-called “social credit” system that

collates disparate sources of data ranging from financial credit scores to criminal records, among others.⁹⁰

Consider, then, the current state of reasoning: China has a population four times larger than the United States, its government began years earlier than the U.S. government to use massive social AI applications, its firms are buying cloud services at a remarkable clip, and yet it is somehow regarded as having a fraction of the usable compute (widely discussed using the metric of data centers) held by the U.S.⁹¹

This defies common sense. And although it reports to the IEA that its data centers are consuming more energy than their U.S. counterparts, the reported numbers still don’t seem to take into account the relatively advanced digital footprint of the Chinese population, companies, and state. China is more likely to have roughly equal or more gross compute (inclusive of cloud services)—albeit dispersed in different types of deployments and counted via different metrics—than the U.S., rather than less.

Estimated data centre electricity consumption and its share in total electricity demand in selected regions in 2022 and 2026



IEA. CC BY 4.0.

Sources: IEA, [Data Centres and Data Transmission Networks](#); Lawrence Berkeley National Laboratory, [United States Data Center Energy Usage Report](#); Ireland Central Statistics Office, [Data Centres Metered Electricity Consumption 2022](#); Danish Energy Agency, [Denmark's Energy and Climate Outlook 2018](#); China's State Council, [Green data centres in focus](#); European Commission, [Energy-efficient Cloud Computing Technologies and Policies for an Eco-friendly Cloud Market](#); Joule (2023), Alex de Vries, [The growing energy footprint of artificial intelligence](#); and Crypto Carbon Ratings Institute, [Indices](#).

With respect to supercomputers, China has been investing in the development of as many as 10 exascale computer systems over the past two years and, after years of being opaque regarding its supercomputer performance, announced a goal to exceed 300 exaflops⁹² of aggregate computing power within two years—a 50 percent increase over its ostensible 2023 capacity of 200 exaflops—by this year (2025).⁹³

If achieved and independently verified, this would transform China's status from being nearly invisible in the supercomputer rankings⁹⁴ to being in first place by a wide margin, surpassing the current U.S.-based supercomputer leader, El Capitan.⁹⁵

The races for artificial general intelligence (AGI) and artificial superintelligence (ASI) have topped headlines for several years and continue to be regarded, rightfully so, as threshold events in the history of humanity. While such metrics are beyond the scope of this paper, it is worth observing a “Field of Dreams” caveat: the relative paucity of debate thus far about practical bounding factors to optimization, governance, B2C use cases, reliable forecasting for ongoing development, monetization, and market valuation.⁹⁶

Critical Factors Are Soon Likely To Favor China

Unlike other bilateral technological races throughout history, say, the U.S.-Germany race for the first atomic bomb or the U.S.-Soviet race to the moon, there is not—indeed, never will be—a single agreed finish line for any type of artificial intelligence “race”.

AI represents a complex and dynamic collection of ideas, talent, funding, raw factors of production, manufacturing, coding, available data, use cases, governance, finance, innovation, geopolitical and commercial alliances, standard-setting, performance benchmarking, and, at times, sheer luck in experimentation. However much the United States has held a commanding position historically, the current dynamic no longer allows for an ordinal ranking of a “first place” finisher, gold medal winner, etc.

Accordingly, independent AI sector analysts have tended to evaluate the U.S.-China competition via multiple factors that show the relative robustness, trajectories, and constraints of the two nations’ AI wide-ranging ecosystems.⁹⁷

While beneficial for context, such expansive assessments have an inherent drawback. Sifting out priorities among dozens of issues becomes challenging for policymakers and nearly impossible for merely fluent stakeholders.

This paper proposes an alternative, perhaps more focused and helpful, approach. It asks: in the evolving AI competition between the United States and China, which metrics are critical determinants that would allow sustained preeminence? Put another, within the next 12 to 18 months, which nation is more likely to be in a better position to control its AI destiny?

The so-called race for AGI and ASI, while consequential, will not be fully determinative of each nation’s relative market position, at least not in the short run. If a considerable portion of startup AI companies are currently struggling to monetize generative models across a broad spectrum of applications—and the use cases for AGI and ASI are enthralling but still speculative—it is hard to envision a scenario in which the economies of both nations could instantaneously monetize models that are harder to design, more expensive to operate, harder to control, harder to predict, and harder to maintain.

Rather, we have entered an era in which there is already a critical mass of frontier AI models provided by existing players that are sufficient—“not perfect, but good

enough”—for most practical, commercially viable applications. The stunning advances in areas such as protein folding, DNA mapping, and molecular design of therapeutic drugs have yielded order-of-magnitude gains in potential that will take companies years, if not decades, to fully develop, much less test, refine, obtain regulatory approval for, and monetize.

**If there is one key metric, it is this:
which nation will be in a better
position to control its AI destiny?**

In plain words, the AI race among research leaders will go on and will be fascinating to track, but the distance between the leaders and

the formidable competitors following in the closest peloton will be more a matter of fan commentary than existential survival.

The champion’s mantle of world-beating ASI will be relevant for national security considerations—cybersecurity can be a sudden-death, win-or-die game—but the same logic is not true for diversified, evolving markets of interdependent companies and governments that sustain themselves via a mixture of legacy products, open-source models, extended product development times, and perpetual opportunities for upgrades. Journalists love a reductive “who’s number one” story line; reality is more complex and multidimensional.

What the U.S. Needs Most But Will Not Get Soon

The U.S. AI sector has three glaring deficits that cannot be easily resolved in the near term:

- sufficient electricity to power the AI compute that will be demanded from data centers nor a regulatory environment that allows for rapid increase in generation; and,
- stable availability of rare earth elements for manufacture of advanced semiconductors and related AI products;
- a coherent and efficient political economy strategy capable of rapidly allocating discretionary funding in the form of subsidies and incentives, and resolve lethargic permitting and siting processes, to address the power-data center nexus.

By far, the most concerning deficit in the U.S. artificial intelligence sector is the availability of electricity for data centers that handle processing for AI models.

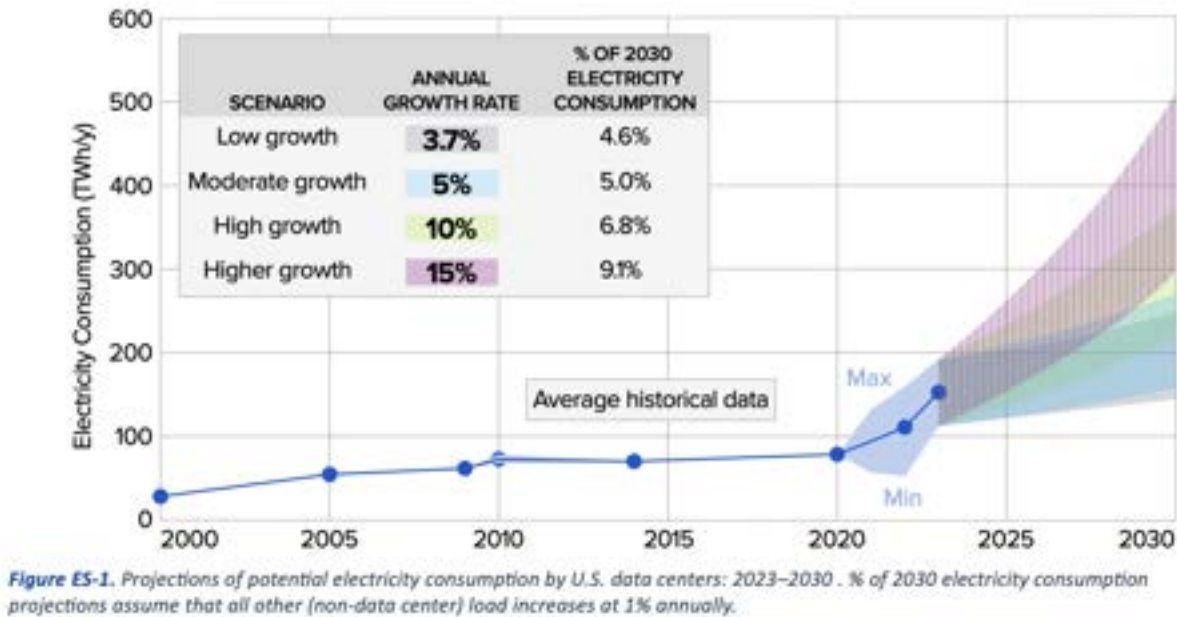
The acceleration of demand for AI processing has caused a data center boom in the United States. In January 2021, there were roughly 8,000 data centers worldwide, roughly one-third of which were located in the U.S. In the span of three years, the number of data centers in the U.S. doubled to more than 5,300.⁹⁸ A more important metric is that data center power demand in 2014 was 16 GW and reached 55-60 GW by the end of 2024.

The U.S. is already struggling in some locations to meet electricity demand driven by the AI sector. Such a challenge is not easily fixed, and it could worsen.

AI-ready data centers have high average power densities, i.e., the energy consumption of servers in the racks. As noted above, estimates of data center power demand vary widely. Mid-range modeling suggests that the current annual

power demand of 60GW by data centers globally is expected to increase at annual rate of 19 to 22%, reaching a range of 171 to 219 GW by the end of this decade. This would represent an unprecedented increase in power demand: “To avoid a deficit, at least twice the data center capacity built since 2000 would have to be built in less than a quarter of the time.”⁹⁹

As of the close of 2024, numerous industry analysts had concluded that, based on current trajectories, the compute demands of generative AI models on data centers may exceed the collective ability of America’s utilities within 12-24 months.¹⁰⁰

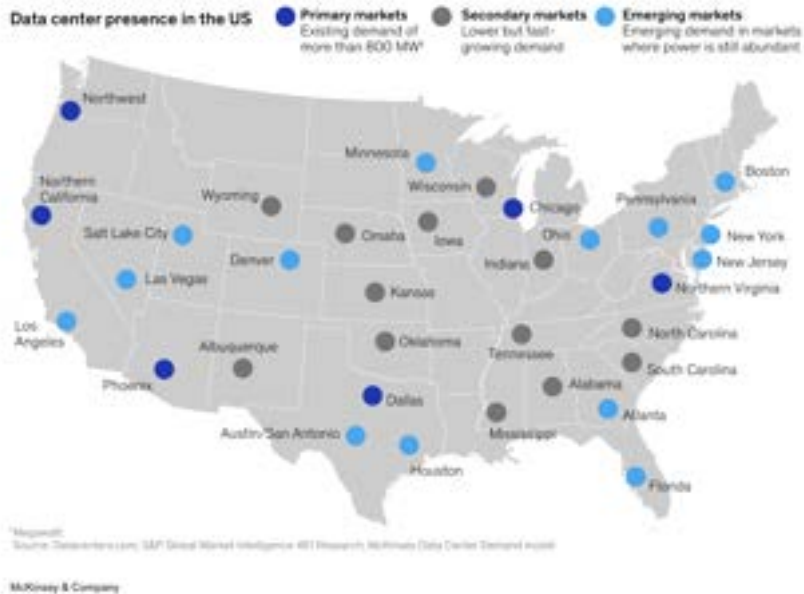


SOURCE: EPRI

Tangible signs of a power squeeze on AI development in the US are already in evidence:

- An analysis by TD Securities finds that the Dallas-area grid can no longer reliably provide power for its local data centers; Northern Virginia may exceed reliability thresholds as soon as 2027; New Albany, Ohio will run out of reliability-rated power in summer 2028; and Silicon Valley will run out of reliability-rated power in summer 2034.¹⁰¹
- The northern Virginia region, sometimes called the data center capital of the world because it represents some two-thirds of global capacity, saw its vacancy rate drop to less than 1 percent in 2024.¹⁰²
- Already, utilities in U.S. regions with significant numbers of data centers are resorting to a variety of tactics to manage surges in AI-driven requests, such as price incentives to encourage scheduling of demand, load-smoothing through the use of additional batteries, upgrading of legacy power generation facilities, addition of renewable energy power generation, and sophisticated software to forecast manage peaks and lulls in demand.¹⁰³
- New data center capacity expected to come online during the next 2-3 years has already been leased out.¹⁰⁴ Estimates are that current capacity utilization rates are at 91% but headed to a peak of 97% - levels that allow no room for unanticipated demand or breakthroughs.
- Data centers in the U.S. that focus on training models, which typically place less emphasis on latency, are increasingly migrating to rural areas where surplus electrical generation capacity still exists and there are fewer regulatory burdens.
- Such is the level of concern across the U.S. federal government that the General Services Administration—citing long-term price concerns and the increasing demand from data centers and AI processing—recently took the unprecedented step of locking in a contract for 10 million megawatt-hours of nuclear-generated power over the next 10 years.¹⁰⁵

Data centers are emerging in more remote locations, where power is still abundant and grids less strained.



Contrast the U.S. scramble for energy efficiency and marginal expansions of ever-smaller facilities with China's amalgam of massive power generation plants. China has roughly half of the largest power generation facilities in the world as ranked by megawatts; the U.S. has one among the top 20 leaderboard.

Relative scale matters also. Among the regions where data centers predominate, the IEA found that U.S. data centers are, on average, using about 6 percent of gross electricity consumption, whereas China's data centers are only using roughly 3 percent (see IEA chart above).¹⁰⁶

Daunting regulatory and political challenges bedevil expansion of energy infrastructure across the United States. In the past five years, about one-third of wind and solar projects were cancelled or experienced delays of six months or more because of zoning disputes, grid connection problems, and local opposition, according to a survey by the Lawrence Berkeley National Laboratory.¹⁰⁷

The study further found that most projects take four to six years from initial announcement to commencement of commercial operations, and about 20 percent of such projects took more than six years. More than 300 counties across the U.S. have banned or instituted moratoria on wind and solar projects.¹⁰⁸

Rather than being mired in a morass of protracted local NIMBY (“not in my backyard”) permitting and environmental disputes, China is a nation that—in the case of a single facility, the Three Gorges Dam hydropower generation plant—was willing to displace an estimated 1.2 million citizens across some 2,000 towns and villages in the Yangtze River region over a 16-year period.¹⁰⁹

In January 2025, Chinese officials confirmed they are working an even more ambitious initiative: a plan to build massive solar power stations in outer space that would produce energy comparable to the Three Gorges Dam and transmit it back to Earth.¹¹⁰

As noted above, China has such a predominant position in terms of REE deposits, processing, and export that it is unlikely to change within the span of a generation. Yet the two biggest rare earth mining companies outside of China, U.S. companies MP Materials and Lynas, “have struggled to expand refining capacity despite huge U.S. government investments. Given the U.S. reliance on China for heavy rare earths used in defense applications, the Department of Defense has provided millions of dollars to both companies and others to fully indigenize ‘mine to magnet’ REE supply chains.”¹¹¹

What China Needs Most And Is Rapidly Obtaining

China has three glaring deficits, also. However, it is proving adept at resolving them and may likely have some of them entirely resolved by the end of 2025:

- a lack of world-leading semiconductor research, design, and foundry capabilities. (China aims to reduce its reliance on imports to 30% by 2025¹¹², and analysts have estimated it will fully close its semiconductor gap with the U.S. by 2027)¹¹³.

- scalable compute capability in the form of existing data processing centers and data center racks;
- an industrial ecosystem that attracts and retains top-tier STEM talent in AI fields from across the globe.

In the U.S., hundreds of projects are being thwarted by “not in my back yard” disputes. China, by contrast, resettled 1.2 million people for a single mega-hydropower project and is working toward a solar-powered ‘Three Gorges Dam’ in outer space.

It is worth recalling here that China has a track record of technologically “leap-frogging” the United States in multiple sectors where AI know-how figures

prominently, such as facial recognition, smart city technologies, mobile payment systems, among others.

China's weak underbelly in AI is its semiconductor industry. It is no surprise, then, that this is precisely the point where the U.S. has chosen to exert further pressure via export controls.

U.S. attempts to enforce its export controls, however, are widely and appropriately regarded as only modestly effective at best; China has negated U.S. attempts to capitalize on its weakness in semiconductors through a variety of legal and other means. As the U.S. was slow to tighten export controls, the Chinese government and Chinese firms accelerated the purchase and stockpiling of advanced chips from the U.S. Such stockpiles are reported to equal several years' worth of demand.¹¹⁴

How much of a head start did China have? It declared the need for semiconductor self-reliance in 2015, nearly three years before the Trump Administration's export

China has used a variety of strategies to successfully blunt the impact of U.S. export controls on AI hardware. controls launched the era of US semiconductor export controls in 2018.¹¹⁵

The Chinese government has ramped up subsidies to its domestic firms to upgrade their capacity, import semiconductor equipment and software, finance the acquisition of foreign semiconductor firms, and reportedly pressured or encouraged U.S. and foreign semiconductor firms to transfer key technology, intellectual property, talent, and R&D to domestic entities.¹¹⁶

FACTORS OF PRODUCTION

RARE EARTHS
WATER
POWER
PERMITTING
GENERATION
COAL
NATURAL GAS
HYDRO
NUCLEAR
RENEWABLES
LARGE-SCALE BATTERY STORAGE
TRANSMISSION
DISTRIBUTION
EFFICIENCY



WHO HAS THE ADVANTAGE?



China has much more control of its destiny with respect to basic factors of production, especially in the areas of rare earth minerals, electrical generation and transmission, and the ability to scale up rapidly

INVESTMENT

PRIVATE
PUBLIC

American private investment is larger and more nimble than that of China, but China's big bets on national champions are paying dividends ...

INTELLECTUAL CAPITAL

GRADUATES
PATENTS
R&D

The U.S. has been a pioneer in intellectual capital, but the sheer amount of Chinese brainpower from its elite universities is closing the gap ...

CHIP FOUNDRY CAPACITY

CONVENTIONAL
QUANTUM

The U.S. has a commanding lead in chip development and market share but ...

COMPUTE

DATA CENTERS
DATA RACKS
TOTAL SUPPLY
PUE

... China's deficit in data racks is less than perceived, and it is getting much of what it needs through cloud services anyway.

MODELS / PARAMETERS

SLURM
LLMs
TRAINING DATA
TEXT
VIDEO
AUDIO
OPEN
CLOSED
AGI
BSI

China placed an early big bet on open-source models that positions itself well and it has enough LLMs near the top of the leaderboards to get the job done for a long time. With its large population (4x the U.S.) and fewer privacy constraints, its data pool for training models is formidable.

APPLICATIONS / SECTOR

AEROSPACE
AGRICULTURE
CHEMICAL
DRONES
EVs
FINANCE
HEALTH CARE
LIFE SCIENCES
MANUFACTURING
MINING / OIL/GAS
RENEWABLES
ROBOTICS
WEAPONS

The U.S. pioneered AI applications across a broad array of industrial sectors. China doubled down on two critical fronts: clean energy, and the use of AI for social aims (policing, education).

QUANTUM

PUBLIC
PRIVATE

It's difficult to predict who will have leadership in quantum computing by the end of this decade.

NOTE: This is the author's "back of envelope" checklist survey only based on public reporting only. A blue square indicates a U.S. leadership position; a red square indicates China; and gray squares are instances in which a lead is marginal or difficult to estimate. This chart is a rough, indicative exercise only, i.e., it is not intended to imply that the metrics should be given equal weight in dollars, units sold, market share, capacity, etc. relative to one another.

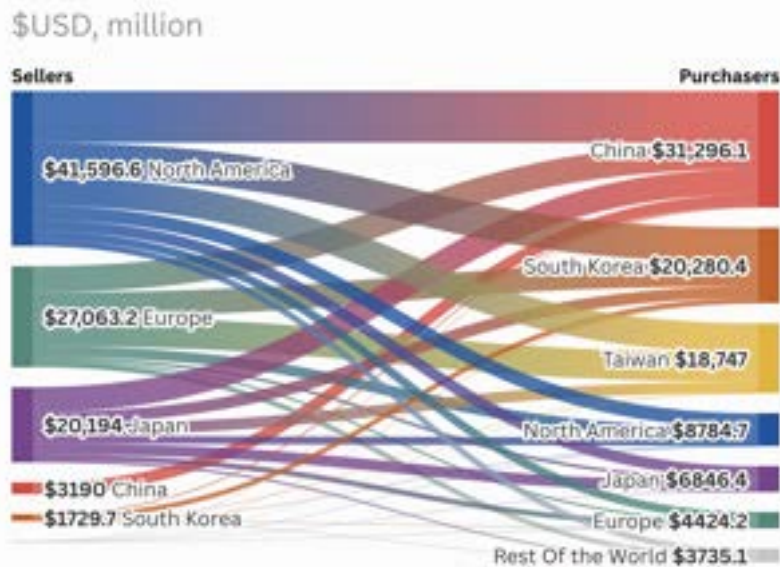
In late 2024, Huawei insiders divulged that—despite production problems with its 910B semiconductor chip in part due to U.S. restrictions on the export of extreme ultraviolet lithography technology—the company would be forging ahead with mass production and sale of the chip early in 2025, giving priority to government and corporate orders. Huawei’s Ascend 910C is meant to be China’s answer to Nvidia’s market-leading chips, such as the A100 and H100.¹¹⁷

Have U.S. export controls been effective? The short answer: not entirely. The U.S. has an estimated four-year lead over China with respect to the price-performance of hardware that trains frontier models but that has not prevented Chinese companies from being serious contenders in the performance of frontier models. The H20 chip that Nvidia developed for the Chinese market with specifications that did not invoke export restrictions has performed well for Chinese customers for the purposes of short-context and long-context inference models, which as of 2025 have been rapidly gaining in popularity.¹¹⁸

Several Chinese startups such as DeepSeek and Moonshot AI (the latter backed by Chinese internet giants Alibaba and Tencent) showed in late December 2024 and early 2025 that their large language models—despite U.S. export sanctions on hardware—were performing favorably, and along some dimensions outperforming, OpenAI’s o1 reasoning model at a tiny fraction of the cost. The unintended consequence of export controls has led to Chinese companies gaining a considerable advantage in making models cheap and efficient without sacrificing performance, a strategy U.S.-based frontier model companies will need to adopt. Furthermore, there is evidence that some Chinese AI developers have accessed the capabilities to restricted Nvidia chips by trading with middlemen and using overseas data centers.¹¹⁹

This whack-a-mole dynamic—wherein China obtains advanced U.S. technological capabilities without initially manufacturing them—seems not unlike the story of America’s global positioning system (GPS). From the inception of GPS in 1973, the United States kept export bans on aerospace and satellite hardware. China, however, was allowed to piggyback on the U.S. global positioning system hardware starting in the early 1990s. This provided China with time to develop its indigenous global positioning system, BeiDou, which took less than a decade to launch in its initial deployment with nationwide coverage. By 2023, the [U.S.] National Advisory Board on Space-Based Positioning Navigation and Timing stated bluntly: “[America’s] GPS capabilities are now substantially inferior to those of China’s BeiDou.”¹²⁰

Figure 1: Global Semiconductor Manufacturing Equipment Supply and Demand by Region, 2023



Note: Supplier region refers to the country headquarters of the seller firm, which is not necessarily the same as the country from which the goods were exported; Buyer region refers to the location of the customer facility in which equipment is installed, whether that is a locally owned or foreign owned firm.
Source: CSIS Analysis; and "Semiconductor Equipment Database," TechInsights.

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AI CENTER

Some Chinese firms, meanwhile, have turned to cloud-computing facilities to access the compute of advanced chips—a move which does not violate export controls as they do not govern direct export or transfer of physical commodities, software, or technology.¹²¹ Last year, the president of AWS Greater China, Chu Ruisong, noted that Amazon’s cloud platform, Bedrock, was providing its clients with “a selection of leading LLMs including prominent closed-source models such as Anthropic’s Claude 3.”¹²² Amazon CEO Andy Jassy told financial analysts in his October 2024 earnings call that, in the previous 18 months, AWS had released about “twice as many machine learning and GenAI features as the other leading cloud providers combined.” Its AI business, he said, has been growing at a “triple digit year-over-year percentage.”¹²³ Just weeks later, Amazon introduced a new family of multimodal generative AI models called Nova, the premier version of which will arrive in early 2025. The company claims they are among the fastest in their class and among the least expensive to operate.¹²⁴

U.S. export controls allow for service contracts to advanced chip manufacturing in China that address facility planning, installation, and repairs.¹²⁵ Moreover, Chinese entities have adopted age-old strategies to work around export restrictions in the artificial intelligence sector: middlemen, disassembling advanced products into smaller (putatively legal) parts; creation of shell and front companies in other nations.¹²⁶

The disparity between the reach of export laws and the nimbleness of the market is such that it is said to take mere days to create and operate a shell company but years to demonstrably prove one is operating outside the law.¹²⁷

One should recall here that wide-ranging use cases and market applications of artificial intelligence are the end-of-the-line litmus test of AI's value. On this front, China has unquestionably excelled. A generation ago, China was behind in batteries, drone technologies, solar energy, and electric vehicles. Today, it has an exceedingly strong market position and, in certain subsectors, a leadership position.

Bottom Lines

If there is any remaining doubt about the seriousness of China's ascent in artificial intelligence—and how closely China is on the heels of America with respect to critical factors as of 2025—one need look no further than the survey of the China AI landscape published in OpenAI's "Infrastructure is Destiny" paper from September 2024¹²⁸, and its "Policy Roadmap" from January 2025¹²⁹. Among the proof points that ring an alarm bell:

- China already counts more than 230 LLMs, more than 180 of which have been approved for public use by the government. Notable GPT-4o level AI models include: Alibaba's Qwen2, 01AI Yi-34b, SenseTime 5.5, WuDao, and Baidu Ernie 4.0. DeepSeek-Coder-V2 outperforms GPT -4 Turbo on evaluation metrics.
- China recently approved 11 nuclear reactors, an investment of at least \$31 billion, with construction expected to take about five years. China has more nuclear reactor under construction than any other nation in the world: 10 new reactors approved per year for each of the last two years.
- China is Number 2 in global compute capacity behind the US.
- China is on target to hit 300 EFLOPS by 2025.

As noted in this paper above, it matters little if China is Number 2 in gross compute capacity if, as a practical matter, its posture allows it to handily best the U.S. in a large and growing number of other metrics nonetheless.

In sum, China relative to the United States enters 2025 with more control over its raw factors of production (rare earth elements), a grid that is more scalable, a formidable and growing amount of intellectual capability, a deficit in compute that is significant but manageable, demonstrable proof that it can keep pace with and on some metrics best U.S. generative AI models, a robust

collaborative and attractive open-source platform regime, an matchable ability to test-bed products and services in a large population, significantly cheaper and more

efficient models, and a convincing case that it efficiently develop use cases and find global markets for products that rely heavily on AI. The U.S., with its lead in compute, private sector financing, world-leading innovations, and wide-ranging dynamism in product development and application has an unassailable claim to a pioneering role, but an AI ecosystem that will continue to contend, for practical, go-to-market purposes, with constraints that are difficult to resolve.

China has proven convincingly that it can keep pace with the U.S. across a broad range of AI metrics.

Instances of “plain talk” in highly technical fields such as AI—while rare, reductive, and often overlooked—can help stakeholders with less technological fluency understand conclusory findings:

“China stands today as a full-spectrum peer competitor of the United States in commercial and national security applications of AI. Beijing is not just trying to master AI—it is succeeding,” according to a report by Harvard scholar Graham Allison and industry leader Eric Schmidt.¹³⁰

“This aggregate metric [of AI accelerators] misidentifies the decisive factor. The real question is whether China can centralize sufficient compute resources to enable at least one domestic AI lab to match U.S. capabilities. National compute totals are not what matter; concentrated capabilities are.”¹³¹

Exogenous and unforeseen factors, admittedly, could rapidly shift the balance of power in the bilateral competition: a trade war that spirals into unintended consequences; a national security crisis or confrontation; a game-changing leap forward in quantum computing by either nation; a bursting of the private sector “bubble” in AI investing; and a major investment on a national champion that proves

ill-advised. However, nations do not customarily rely on a lucky draw as a geopolitical strategy.

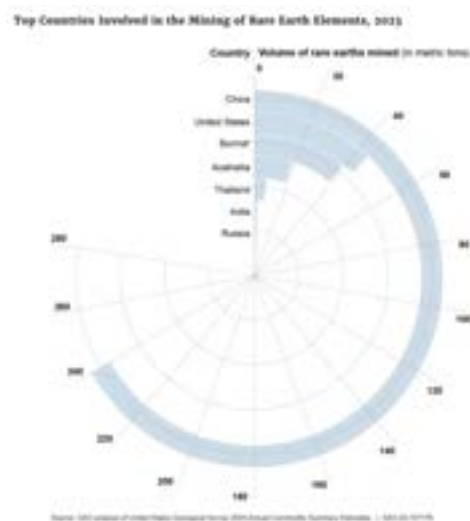
Policy Options for the U.S.

If China is a near-peer or nose ahead as of 2025—or at least in an enviable pole position to control its AI destiny for the remainder of this decade—it is worth exploring the numerous policy implications for the U.S.

In light of the moves made by the outgoing Biden and incoming Trump administrations, the United States appears poised to further expand its whole-of-government approach to maintaining America’s lead. The following is an overview of policy initiatives that may be worth considering.

Major But Not Comprehensive Reforms. Presidential terms—especially those of a term-limited president—do not have unlimited bandwidth for legislative agendas. As such, the agenda of the Trump administration may need to be limited, for practical purposes, to reforms that are major but not comprehensive in scope. Examples of targeted initiatives could include:

- creation of a national stockpile or stockpiles for rare earth elements in order to bolster national security, protect supply chains, and cushion against price shocks. Rare earth elements are fundamental to a wide array of U.S. weapons and defense systems ranging from the F-35 fighter jet to Aegis destroyers to missile guidance systems;¹³²



- issue an executive order directing the U.S. Export-Import Bank and U.S. International Development Finance Corporation to place a priority on infrastructure and mining deals related to rare earth minerals in developing nations;
- creation of a blue-ribbon commission for reform of energy-related permitting at the federal, state, and local levels, perhaps under the auspices of the new Department of Government Efficiency (DOGE). Such a commission would be given a firm deadline for proposing recommendations to the administration and Congress;
- Endorse and propose a rapid adoption of major legislation conceived in the Senate roadmap for AI¹³³;
- reorganize disparate elements of the AI governance community into a new “Institute for Artificial Intelligence”, which would be charged with research, standard-setting, agenda-setting, informing legislators on policy options, etc. much like the mandate of the National Institute of Health for the medical and healthcare sector. The 2023 Biden Administration executive order on AI indicates that federal AI governance is currently spread across 50 different agencies; the order encompassed 150 directives.¹³⁴; and,
- Under the auspices of a national security imperative or emergency declaration, propose an elimination of H-1B visa caps for a select class of AI-related fields, blanket forgiveness of student loans in such sectors, and free tuition for qualified students pursuing education in such sectors.

A More Ambitious Agenda. A more ambitious agenda could see the incoming Trump administration pursue a comprehensive, multi-pronged strategy that combines domestic and international elements. American industry could collaborate even more closely with USG and begin to take steps to improve its overall resilience in the global marketplace. Among the potential elements:

- creation of a National Infrastructure Bank to address, in part, deficits in the nation’s energy infrastructure.¹³⁵ This idea has been gaining ground in recent years and legislation has been introduced. Such a step could revitalize the push for infrastructure build-out in natural gas and battery storage. Conceivably, a National Infrastructure Bank could house the newly announced \$500 billion Stargate fund, whose administrative apparatus is still only loosely defined.

- creation of a new AI agency, as noted in the previous section, along with the various reforms outlined therein. Such an agency might be allocated supervision of a dramatically increased pool of grants and subsidies for the frontier technologies of AI, such as quantum computing;
- strategic investments and industry consolidation in Europe and India, as a counterweight to China. European nations, with comparatively weak domestic energy demand and smaller AI markets, may soon look to latch up with large American firms with deeper pockets for future competition;
- a revisit and upgrade of the CHIPS Act to expand and extend its scope such that a wider array of companies may benefit;
- a robust initiative to promote and fund energy efficiency, which has long been an afterthought and de facto stepchild in U.S. energy policy. This could be elevated to a national priority and subsidies could be directed appropriately;
- a concerted push to assess and promote the availability of large troves of currently inaccessible public data at the federal, state, and local levels, and perhaps troves of data from failed American companies. For example, the federal Pension Benefit Guaranty Corporation—in assuming pension liabilities for failed companies—obtains some access to other company assets through bankruptcy proceedings, negotiated settlements, and liens. Data could become one of the assets regularly put on the table;
- a push to accelerate decentralization of power generation across the U.S. by bolstering support for mini- and off-grid renewable energy facilities; and, further, take steps to allow such companies, where possible, to sell their surplus power to nearby companies or across the grid via “wheeling”;
- reappraising and perhaps waiving anti-trust constraints on a select number of U.S. companies, such that the United States would have a de facto “national champions” industrial policy;
- trade negotiations with China to achieve a permanent rapprochement on issues such as rare earth elements, sharing of training data, and expeditious resolution of the growing bow wave of patent AI disputes, perhaps through the creation of a specialized court under the auspices of the International Centre for the Settlement of Investment Disputes.

- an effort to have China and other nations be reclassified in the World Trade Organization to reduce their “developing nation” preferential treatment.
- propose and develop an American analog of China’s Belt-and-Road Initiative, wherein the U.S. would plan, prioritize, accelerate, and subsidize investment and co-investment within nations and sub-sectors that are clearly advantageous for U.S. industry, e.g., rare earth prospecting, mining, and processing within Africa; within Mexico and Canada, support investments in data centers and power generation facilities that are located near US borders; etc. There is some precedent for this type of coordinated approach in the form of the Power Africa initiative, which was started during the Obama administration;
- a revitalization and expansion of the U.S. State Department’s secretariat for Public Diplomacy and Public Affairs. The secretariat would be charged with encouraging developing nations to adopt English as an official language and incorporate such a standard into their AI purchasing protocols;
- Evaluate and reform the mandate of the U.S. Federal Reserve to increase its collaboration with other central banks on AI issues (much as it has been doing on climate change initiatives) and extend targeted outreach to ensure sovereign wealth funds are making long-term investment and standards decisions that are favorable to US economic stability and industrial interests. (Alternatively, the latter goal might be pursued by the Department of Commerce or newly created Institute of Artificial Intelligence).

Final Thoughts

Whether in terms of geopolitical balance of power or positioning of companies in global markets, AI is now firmly established as the apex technology and critical determinant of national autonomy among the U.S., China, and rising powers. As of 2025, China is now highly likely to have closed—or soon close—gap with the U.S. and may well be in more firm control of its destiny.

China has proven that its ability to take a long-term strategic view of its industrial development can yield results that are as rapid and consequential of those of America’s decentralized, market-centric approach. A fundamental reassessment and reframing of the terms of debate in the bilateral dynamic is overdue.

China has likely closed the gap, or soon will, with the U.S. in AI. A comprehensive reassessment is overdue.

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